

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

|                      |                         |
|----------------------|-------------------------|
| Perihalan Produk:    | <u>Calcium granules</u> |
| Product Description: | <u>Calcium granules</u> |
| Cat No. :            | 00875                   |
| Synonyms             | Calcium Metal.; Calciat |
| CAS No               | 7440-70-2               |
| Molecular Formula    | Ca                      |

**Relevant identified uses of the substance or mixture and uses advised against**

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

|                |                             |
|----------------|-----------------------------|
| E-mail address | Enquiry.my@thermofisher.com |
|----------------|-----------------------------|

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                                        |                   |
|------------------------------------------------------------------------|-------------------|
| Substances/mixtures which, in contact with water, emit flammable gases | Category 2 (H261) |
|------------------------------------------------------------------------|-------------------|

**Label Elements**

**Signal Word**
**Danger**
**Hazard Statements**

H261 - In contact with water releases flammable gases

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## Precautionary Statements

### Prevention

P231 + P232 - Handle and store contents under inert gas. Protect from moisture

P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water

### Storage

P402 + P404 - Store in a dry place. Store in a closed container

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Calcium   | 7440-70-2 | 99       |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention if symptoms occur.

#### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention if symptoms occur.

#### Ingestion

Clean mouth with water. Get medical attention.

#### Inhalation

Remove from exposure, lie down. Remove to fresh air.

#### Self-Protection of the First Aider

No special precautions required.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

#### Extinguishing media which must not be used for safety reasons

Water.

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## Special hazards arising from the substance or mixture

Water reactive. Produce flammable gases on contact with water.

## **Hazardous Combustion Products**

Hydrogen, Calcium oxides.

## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation.

### Environmental precautions

See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Do not expose spill to water. Do not let this chemical enter the environment.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Do not allow contact with water. Avoid contact with skin and eyes. Avoid contact with skin and clothing. Do not ingest. If swallowed then seek immediate medical assistance. Protect from moisture.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Protect from moisture. Keep from any possible contact with water.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### **Engineering Measures**

None under normal use conditions.

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## Personal protective equipment

|                                 |                                                                          |
|---------------------------------|--------------------------------------------------------------------------|
| <b>Eye Protection</b>           | Goggles                                                                  |
| <b>Hand Protection</b>          | Protective gloves                                                        |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Respiratory Protection</b>   | No special protective equipment required |
| <b>Recommended Filter type:</b> | Particulates filter conforming to EN 143 |

|                                |                                                                       |
|--------------------------------|-----------------------------------------------------------------------|
| <u><b>Hygiene Measures</b></u> | Handle in accordance with good industrial hygiene and safety practice |
|--------------------------------|-----------------------------------------------------------------------|

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <u><b>Environmental exposure controls</b></u> | No information available |
|-----------------------------------------------|--------------------------|

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                     |                          |                                          |
|-------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                   | Grey                     |                                          |
| <b>Physical State</b>               | Solid                    |                                          |
| <b>Odor</b>                         | Odorless                 |                                          |
| <b>Odor Threshold</b>               | No data available        |                                          |
| <b>pH</b>                           | 14                       |                                          |
| <b>Melting Point/Range</b>          | 845 °C / 1553 °F         |                                          |
| <b>Softening Point</b>              | No data available        |                                          |
| <b>Boiling Point/Range</b>          | 1484 °C / 2703.2 °F      | @ 760 mmHg                               |
| <b>Flash Point</b>                  | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>     | No information available |                                          |
| <b>Explosion Limits</b>             | No data available        |                                          |
| <b>Vapor Pressure</b>               | No data available        |                                          |
| <b>Vapor Density</b>                | Not applicable           | Solid                                    |
| <b>Specific Gravity / Density</b>   | No data available        |                                          |
| <b>Bulk Density</b>                 | No data available        |                                          |
| <b>Water Solubility</b>             | Reacts with water        |                                          |
| <b>Solubility in other solvents</b> | No information available |                                          |

**Partition Coefficient (n-octanol/water)**

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|                           |                          |       |
|---------------------------|--------------------------|-------|
| Autoignition Temperature  | No data available        |       |
| Decomposition Temperature | No data available        |       |
| Viscosity                 | Not applicable           | Solid |
| Explosive Properties      | No information available |       |
| Oxidizing Properties      | No information available |       |

|                   |       |
|-------------------|-------|
| Molecular Formula | Ca    |
| Molecular Weight  | 40.07 |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes.  
Reacts violently with water.

### Chemical Stability

Moisture sensitive.

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

### Conditions to Avoid

Incompatible products. Exposure to moist air or water.

### Incompatible Materials

Acids. Strong oxidizing agents. Alcohols. Ammonia. Halogens. oxygen.

### Hazardous Decomposition Products

Hydrogen. Calcium oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

|                     |                                                             |
|---------------------|-------------------------------------------------------------|
| Product Information | No acute toxicity information is available for this product |
|---------------------|-------------------------------------------------------------|

|                     |                   |
|---------------------|-------------------|
| (a) acute toxicity; |                   |
| Oral                | No data available |
| Dermal              | No data available |
| Inhalation          | No data available |

|                                |                   |
|--------------------------------|-------------------|
| (b) skin corrosion/irritation; | No data available |
|--------------------------------|-------------------|

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(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available

Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Reacts with water so no ecotoxicity data for the substance is available. Do not empty into drains.

### Persistence and degradability

Persistence Persistence is unlikely, based on information available.  
Degradability Not relevant for inorganic substances, Reacts with water.  
Degradation in sewage Water reactive.  
treatment plant

Bioaccumulative potential Product does not bioaccumulate due to reaction with water

Mobility in soil Reacts with water. Is not likely mobile in the environment.

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Other adverse effects No information available

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## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### Other Information

Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Solutions with high pH-value must be neutralized before discharge

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1401  
Hazard Class 4.3  
Packing Group II  
Proper Shipping Name CALCIUM

### Road and Rail Transport

UN-No UN1401  
Hazard Class 4.3  
Packing Group II  
Proper Shipping Name CALCIUM

### IATA

UN-No UN1401  
Hazard Class 4.3  
Packing Group II  
Proper Shipping Name CALCIUM

#### Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

X = listed

| Component | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------|-----------|------|-----|-------|------|------|-------|------|----------|
| Calcium   | 231-179-5 | X    | X   | X     | X    |      | X     | X    | KE-04462 |

### National Regulations

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

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## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

**Revision Date**

**Revision Summary**

Health, Safety and Environmental Department

25-Mar-2025

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**